

COIMBATORE INSTITUTE OF TECHNOLOGY, COIMBATORE-14.

DEPARTMENT OF INFORMATION TECHNOLOGY

Next-Gen Full Stack Web Development with MERN Stack

15.5.2026 to 28.5.2026

Assignment-3

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TOPIC: Student Management REST API using Node.js and Express.js

GITHUB REPOSITORY LINK:

<https://github.com/Madhavan23-byte/smart-queue-slot-booking-system>

API BASE URL:

<http://localhost:3000>

Submitted Components:

Node.js server, Express.js server, custom middleware to display request method, GET API, POST API, PUT API, DELETE API, complete CRUD operations using an array, and Postman testing screenshots.

API Endpoints Implemented:

- GET /students - Fetch all students
- POST /students - Add a new student
- PUT /students/:id - Update student details
- DELETE /students/:id - Delete a student

Screenshot 1: GET /students - Fetch All Students

The screenshot displays the Postman application interface. The top navigation bar includes the workspace name "Madhavan's Workspace", a search bar, and buttons for "Invite" and "Upgrade". The left sidebar shows the "COLLECTIONS" section with "New Collection" and "GET New Request". The main workspace area is titled "New Collection > New Request" and shows a GET request to "http://localhost:3000/students". The "Params" tab is active, showing a table with columns "Key", "Value", "Description", and "Bulk Edit". The "Body" tab is also active, showing a JSON response with a status of "200 OK", a response time of "32 ms", and a size of "504 B". The JSON response is as follows:

```
1 {
2   "message": "Students fetched successfully",
3   "data": [
4     {
5       "id": 1,
6       "name": "Madhavan ",
7       "rollNo": "2403717610421101",
8       "department": "Computer Science",
9       "year": "III Year"
10    },
11    {
12      "id": 2,
13      "name": "Kanish",
14      "rollNo": "2403717610421102",
15      "department": "Information Technology",
16      "year": "III Year"
17    }
18  ]
19 }
```

The right sidebar contains a "New Chat" section with a message: "Start using Agent Mode! Your plan includes 50 AI credits per month to use Agent Mode." Below this is a "Shape what's next" survey prompt: "Tell us which features matter most to you. Takes under 1 minute." with buttons for "Not now" and "Take the survey".

Screenshot 2: POST /students - Add New Student

The screenshot displays the Postman application interface. The top bar shows the user is in 'Madhavan's Workspace'. The main workspace is divided into three panes. The left pane shows the 'COLLECTIONS' sidebar with 'New Collection' and 'GET New Request'. The middle pane shows the 'New Request' editor for a POST request to 'http://localhost:3000/students'. The 'Body' tab is selected, showing a JSON payload:

```
{  "name": "Priya S",  "rollNo": "2403717610421103",  "department": "Information Technology",  "year": "III Year"}
```

. The right pane shows the 'New Chat' section with a message about AI credits. A 'Shape what's next' survey popup is visible in the bottom right corner.

Request Details:

- Method: POST
- URL: http://localhost:3000/students
- Body (JSON):

```
{  "name": "Priya S",  "rollNo": "2403717610421103",  "department": "Information Technology",  "year": "III Year"}
```

Response Details:

- Status: 201 Created
- Time: 79 ms
- Size: 398 B
- Body (JSON):

```
{  "message": "Student added successfully",  "data": {    "id": 3,    "name": "Priya S",    "rollNo": "2403717610421103",    "department": "Information Technology",    "year": "III Year"  }}
```

Survey Popup:

Shape what's next

Tell us which features matter most to you. Takes under 1 minute.

[Not now](#) [Take the survey](#)

Screenshot 3: PUT /students/:id - Update Student Details

The screenshot displays a REST client interface with the following components:

- Top Bar:** Includes navigation icons, a search bar, and buttons for 'Invite' and 'Upgrade'.
- Left Sidebar:** Contains a 'COLLECTIONS' section with 'New Collection' and 'New Request' (highlighted in green), and an 'ENVIRONMENTS' section with 'SPECS' and 'FLOWS'.
- Main Panel:**
 - Request Section:** Shows a PUT request to `http://localhost:3000/students/1`. The 'Body' tab is active, displaying a JSON payload:

```
1 {
2   "name": "Shushi Anush",
3   "department": "Computer Science",
4   "year": "IV Year"
5 }
```
 - Response Section:** Shows a successful response with status `200 OK`, `9 ms`, and `393 B`. The response body is displayed in JSON format:

```
1 {
2   "message": "Student updated successfully",
3   "data": {
4     "id": 1,
5     "name": "Shushi Anush",
6     "rollNo": "2403717610421101",
7     "department": "Computer Science",
8     "year": "IV Year"
9   }
10 }
```
- Right Sidebar:** Features a 'New Chat' section with a message: 'Start using Agent Mode! Your plan includes 50 AI credits per month to use Agent Mode.'
- Bottom Right:** A survey pop-up titled 'Shape what's next' with the text 'Tell us which features matter most to you. Takes under 1 minute.' and buttons for 'Not now' and 'Take the survey'.

Screenshot 4: DELETE /students/:id - Delete Student

The screenshot displays the Postman interface for a REST client. The top navigation bar includes a search bar, 'Invite' and 'Upgrade' buttons, and notification icons. The left sidebar shows the 'COLLECTIONS' panel with 'New Collection' and 'GET New Request'. The main workspace is titled 'New Collection > New Request' and shows a 'DELETE' request to 'http://localhost:3000/students/2'. The 'Body' tab is selected, showing a 'raw' JSON body with the value '1'. The response is a '200 OK' status with a 7 ms response time and 506 B of data. The response body is displayed in a JSON format, showing a success message and a list of student data.

Request Details:

- Method: DELETE
- URL: http://localhost:3000/students/2
- Body: raw JSON, value: 1

Response Details:

- Status: 200 OK
- Time: 7 ms
- Size: 506 B

Response Body (JSON):

```
1 {
2   "message": "Student deleted successfully",
3   "data": [
4     {
5       "id": 1,
6       "name": "Shushi Anush",
7       "rollNo": "2403717610421101",
8       "department": "Computer Science",
9       "year": "IV Year"
10    },
11    {
12      "id": 3,
13      "name": "Priya S",
14      "rollNo": "2403717610421103",
15      "department": "Information Technology",
16      "year": "III Year"
17    }
18  ]
19 }
```

Notifications:

- New Chat:** Start using Agent Mode! Your plan includes 50 AI credits per month to use Agent Mode.
- Shape what's next:** Tell us which features matter most to you. Takes under 1 minute. [Not now] [Take the survey]

Footer: Connect Git, Console, Terminal, 0 errors, 0 warnings. Globals, Vault, Tools.

Screenshot 5: VS Code Terminal - Server Running and Middleware Output

```
PS E:\IT dept intern\student-management-api> npm start
```

```
> student-management-api@1.0.0 start
```

```
> node server.js
```

```
Server is running on http://localhost:3000
```

```
Request Method: GET
```

```
Request Method: GET
```

```
Request Method: GET
```

```
Request Method: GET
```

```
Request Method: POST
```

```
Request Method: PUT
```

```
Request Method: DELETE
```